Reg. No. | | | | | | | | | | | [18MTEST201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE – 2019

EMBEDDED OS AND RTOS

Time: 03 Hours **Maximum Marks: 100**

Instructions:
1. Answer **FIVE** full Questions. Question1 and Question2 are compulsory.
2. Missing data may be suitably assumed.

1.	**a.**	List and explain the services provided by an operating system.	**10**
	b.	Explain the advantages of the layered approach with a neat diagram.	**10**
2.	**a.**	Briefly explain the preemptive and round robin scheduling algorithms.	**10**
	b.	With a neat FSM explain ready, blocked and running task states scheduling.	**10**
3.	**a.**	With a neat diagram explain semaphore and its typical operation.	**10**
	b.	Explain the message queue with a neat diagram and its typical operation.	**10**
		OR	
4.	**a.**	With a neat diagram discuss event register and control blocks.	**10**
	b.	Write short note on TCP/IP protocol stack and command shell.	**10**
5.	**a.**	Explain classification of general exceptions with general exception priorities.	**10**
	b	Write a short note on exception handler with respect to frames timing and interrupt processing.	**10**
		OR	
6.	**a.**	Write a short note on programmable interval timers.	**10**
	b.	Explain timing wheels with timeout event handler.	**10**
7.	**a.**	What are the components and design principles of Linux OS? Explain.	**10**
	b.	Write short notes on network structure and portability issues in Linux.	**10**
		OR	
8.	**a.**	Explain the components that kernel module support under Linux.	**12**
	b.	Explain how a process is managed in Linux platform.	**08**

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